

Operation and Maintenance Manual

Frac Pack

EPA/CARB

TF12V4000C1

TF12V4000C2

TF12V4000C3

Application Group 4D

SS180039/00E



Power. Passion. Partnership.

System designation	Engine model	kW/cyl.	Application group
TF12V4000C1	12V4000T95R	140 kW/cyl.	4D, Frac operation (continuous operation with low load)
TF12V4000C2	12V4000T95	155 kW/cyl.	4D, Frac operation (continuous operation with low load)
TF12V4000C3	12V4000T95L	162 kW/cyl.	4D, Frac operation (continuous operation with low load)

Table 1: Applicability

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1 Safety

1.1 General conditions

General

Safety notifications (DANGER, WARNING, and CAUTION) are used in this manual to alert the owner and/or operator to special instructions concerning a particular procedure that may be hazardous if performed incorrectly.

In addition to understanding the instructions in this publication, the user is also responsible for ensuring compliance with local safety requirements and regulations.

The safety information contained in this section relates to the engine-transmission package described in this publication and covers:

- Types of safety notifications used in this publication
- Prerequisites for working on an engine-transmission package
- Working on an engine-transmission package
- Unauthorized modifications, add-on devices, and replacement parts

Types of Safety Notifications Used in this Publication

Safety notifications (DANGER, WARNING, and CAUTION) are used in this manual to alert all individuals to special instructions concerning a particular procedure that may pose a hazard if performed incorrectly. These safety notifications alone cannot eliminate the hazards that they signal. Strict compliance with these special instructions and common sense operation are critical accident prevention measures. All safety notifications found on the equipment must be observed. Ensure that safety-related labels are legible and not obstructed by dirt, grease, or other equipment.

DANGER



In the event of immediate danger.

Consequences: Death or serious injury

- Remedial action

WARNING



In the event of potentially dangerous situations.

Consequences: Death or serious injury

- Remedial action

CAUTION



In the event of hazardous situations.

Consequences: Minor or moderate injuries

- Remedial action

NOTICE



In the event of a situation involving potentially adverse effects on the product.

Consequences: Material damage.

- Remedial action
- Additional product information

NOTICE



Manufacturers' recommendations

- All manufacturers' safety notifications are to be followed regardless of format.

The engine-transmission package manufacturer cannot anticipate every possible circumstance that might involve a potential hazard. The warnings in this publication and on the product are, therefore, not all inclusive. If a tool, procedure, work method, or operating technique that is not specifically recommended by the engine-transmission package manufacturer is used, you must satisfy yourself that it is safe for you and for others. You should also ensure that the product will not be damaged or made unsafe by the transport, installation, commissioning, operation, lubrication, maintenance, repair, or storage procedure that is chosen.

The information, specifications, and illustrations in this manual are based on information available at the time it was written. The information, specifications, and illustrations can change at any time. These changes can affect the service given to the product. Obtain the complete and most current information before you start any work on the engine-transmission package.

Prerequisites for working on an engine-transmission package

For the remainder of this publication, the term “working”, “work” or “all work” is used in reference to work done on the engine-transmission package. “Work” or “all work” is defined as transport, installation, commissioning, operation, maintenance, and repair.

NOTICE 	Persons performing work on equipment Equipment damage • Every person performing work on the equipment has a responsibility to read all cautions and become acquainted with symbols before operating or repairing this product.
NOTICE 	Permission for working on the equipment • Always obtain permission from the person in charge before commencing work on the equipment.

Working on an engine-transmission package

Do not work on this engine-transmission package unless you have read and understand the instructions and safety notifications contained in this publication. Failure to follow the instructions or heed the safety notifications could result in injury or death. Contact an authorized sales or service outlet for a replacement manual.

Correct use

The engine-transmission package is intended and designed exclusively for the application, Frac, 4D specified in the contract or defined at the time of delivery. Any other use is considered improper use. The engine-transmission package is to be used in accordance with the manufacturer's instructions as specified in this manual. Correct use includes, but is not limited to, observation of and compliance with the maintenance instructions and observation of maintenance and safety requirements. The engine-transmission package manufacturer will not be liable for any losses and/or damages caused by improper use or actions. The engine-transmission package manufacturer will not be liable for any losses and/or damages caused by improper use or actions resulting from personnel while under the influence of alcohol, medication, or any other mind altering substances.

Hazardous environment operation

The engine-transmission package is not approved for hazardous environment operation. It is the responsibility of the operator to ensure the engine-transmission package is not operated in a hazardous environment. A hazardous environment is defined as a mixture of volatile substances with air, in the form of gases, vapors, mist, or dust in which, ignition can occur.

ATEX in the EU and OSHA in the U.S., define and classifies hazardous environments.

Unauthorized modifications and third-party add-on devices, and replacement parts

NOTICE 	Replacement safety labels • Please contact an authorized service outlet to determine how to obtain replacement labels for parts that have safety labels.
NOTICE 	Manufacturers' safety notifications • All manufacturers' safety notifications are to be followed. • Unauthorized modifications, third-party add-on devices and replacement parts may compromise the integrity of the equipment, and may present a safety risk.

Only genuine spare parts from the engine-transmission package manufacturer must be used to replace components or assemblies.

The engine-transmission package manufacturer accepts no liability or warranty claims for any damage caused by unauthorized modifications, third-party add-on devices, and/or replacement parts. If damage is caused by the use of non-genuine replacement parts, no liability or warranty claims through the engine-transmission package manufacturer will be accepted.

1.2 Personnel and organizational requirements

Publication notice

For the remainder of this publication, the term “work”, “working” or “all work” is used in reference to work done on the engine-transmission package. “Work”, “working” or “all work” is defined as one or more of the following actions: transport, installation, commissioning, operation, maintenance, repair, storage.

For the remainder of this publication, the term “certified personnel” refers to a certified technician who has successfully completed training at the manufacturer's certified training center and who has also been instructed on the engine-transmission package work by means of this publication, with special attention paid to the safety notifications. The manufacturer's factory certified technician training classes are available in various locations to meet worldwide needs.

For the remainder of this publication, the term “operator” or “third-party user” is generally the person involved with the transport, installation, commissioning, operation, maintenance, or repair that does not require the technician to be certified by the engine-transmission package manufacturer, but who must read and have full understanding of this publication, especially the safety notifications, before proceeding with any operation.

For the remainder of this publication the term “owner” or “person responsible for the engine-transmission package operation” is the person who either owns the piece of equipment or facility within which it resides. This person generally schedules or directs the work being done on or with the engine-transmission package.

For the remainder of the publication, “third-party” or “add-on” generally refers to items not designed by or purchased from the engine-transmission package manufacturer; whereas, “genuine” or “factory” means it was designed by or purchased from the engine-transmission package manufacturer. Generally, the use of third-party add-ons and non-genuine replacement parts is the sole responsibility of the owner.

In general, operational checks can be performed by non-certified personnel following the published instructions. All other work on the engine-transmission package must be carried out by properly certified personnel. For the safety of the certified personnel and all others in the vicinity of work being done on the engine-transmission package, it is imperative that this publication is provided to all certified personnel involved in any or all work done on the engine-transmission package.

NOTICE



Manufacturers' recommendations

- All manufacturers' safety notifications are to be followed regardless of format.

Personnel and instructional requirements

This publication must be issued to all personnel involved in the transport, operation, maintenance, repair and/or storage of an engine-transmission package.

Responsibilities of the operating, maintenance, and repair personnel must be specified by the “owner”.

This publication must be kept near the engine-transmission package and accessible to all personnel involved in operation, maintenance, repair and/or transportation.

The personnel must be instructed on engine-transmission package operation and repair by means of this publication, and in particular, the safety notifications must be explained and understood.

Owners are responsible for instructing personnel who only occasionally work on this engine-transmission package, before requesting the work be performed.

NOTICE



Additional safety standards

- Additional national safety standards exist. The manufacturer recommends owners, operators and certified technicians become familiar with national safety standards, where available.
- Operators should refer to the safety requirements for operators in this publication.

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Maintenance safety

Additionally, all personnel working on the engine-transmission package must ensure that all safety notifications and informational messages are read and understood before any work is performed on the engine-transmission package:

- Strictly adhere to the maintenance and repair schedule.
- Never carry out the maintenance and/or the repair schedule with the engine and/or engine-transmission package running, unless specified in the procedure.
- Ensure persons not involved in the work on the engine-transmission package keep clear.
- Never attempt to rectify faults or carry out repairs without the necessary certification and required special tools. For clarification or assistance, contact the nearest authorized service outlet.
- Use proper, calibrated tools.
- Never work on the engine-transmission package, engine, or other engine-transmission package components without following proper lifting requirements.
- Ensure nobody is standing in the danger zone before barring over the engine. After completing the work on the engine-transmission package, check that all protective devices/safety guards have been installed and that all tools and loose parts have been removed from the engine-transmission package.
- Observe special cleanliness when conducting maintenance and/or repair work on the engine-transmission package plant. After completion of maintenance and repair work, make sure that no loose objects are in/on the engine-transmission package.

1.3 Safety regulations for maintenance and repair work

Safety regulations before maintenance and repair work

Have maintenance or repair work carried out by qualified and authorized personnel only.

Allow the product to cool down to less than 50 °C (122 °F) (risk of explosion from oil vapors, risk of burning from fluids and lubricants).

Relieve pressure in fluid and lubricant systems and compressed-air lines which are to be opened. Use suitable containers of adequate capacity to catch fluids and lubricants.

When changing the oil or working on the fuel system, ensure that the service room is adequately ventilated.

Never carry out maintenance and repair work with the product in operation, unless:

- It is expressly permitted to do so following a written procedure.
- The product is running in the low load range and only for as long as necessary.

Lock-out the product to preclude undesired starting, for example, start interlock.

Attach "Do not operate" sign in the operating area or to control equipment.

Hydraulic starting systems are highly pressurized. For additional safety, before working on the engine or system, pressure must be removed from the starter accumulator system according to the manufacturer's specifications. Hang a "Do Not Start" sign on the accumulator system. Recharge the accumulator system after maintenance work has been completed.

Close the main valve on the compressed-air system and vent the compressed-air line when pneumatic starters are fitted. Hang a "Do Not Start" sign on the main valve. Keep the main valve closed until maintenance work has been completed for additional safety.

Disconnect the battery. Lock contacts.

Disconnect the control equipment from the product.

The following applies to starters with copper-beryllium alloy pinions:

- Wear a respirator mask (filter class P3). Do not blow out the interior of the flywheel housing or the starter with compressed air. Clean the flywheel housing inside with a class H dust extraction device.
- Observe the safety data sheet.

Safety regulations during maintenance and repair work

Take special care when removing ventilation or plug screws from the product. Cover the screw or plug with a rag to prevent fluids escaping under pressure.

Take care when draining hot fluids and lubricants (risk of burning).

Use only proper and calibrated tools. Observe the specified tightening torques during assembly or disassembly.

Carry out work only on assemblies or plants which are properly secured.

Never use lines for climbing.

Keep fuel injection lines and connections clean.

Always seal connections with caps or covers if a line is removed or opened.

Fit new seals when reinstalling lines.

Avoid damaging lines, particularly the fuel lines.

Ensure that all retainers and dampers are installed correctly.

Ensure that all fuel injection and pressurized oil lines are installed with enough clearance to prevent contact with other components. Do not place fuel or oil lines near hot components.

Do not touch elastomeric seals (for example, Viton sealing rings) with your bare hands if they have a carbonized or resinous appearance.